IRS Assisted Orthogonal Frequency Division Multiple Access (OFDMA) For Rate Optimization

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Objective and methodology used

AIM: To maximize the common (minimum) rate of users by optimizing the

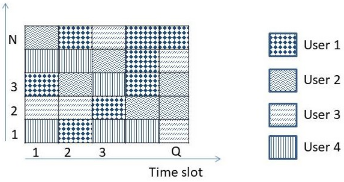
- phase shift matrix at IRS
- OFDMA RB
- power allocation

IRS lacks frequency selective processing
Design a common set of reflection coefficients adapting to large number of channels (of multiple users over multiple sub bands)
Dynamic passive beamforming:

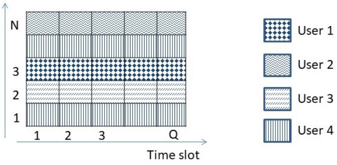
- Different IRS coefficients are set for different time slots
- Small number of users selected in each time slot

Phase is configured at every time slot: $\phi_q = [\phi_{q,1}, \phi_{q,2}, \cdots, \phi_{q,N}]^T \in \mathbb{C}^{N \times 1}$

Same reflection coefficients $\implies$ All RBs tend to be allocated to same user



(a) Dynamic beamforming



(b) Passive beamforming

Figure: Illustration of RBs in dynamic and passive beamforming

Quasi-static block fading channel model

System model

Users: $\mathcal{K} = \{1, 2, \cdots, K\}$, IRS: $\mathcal{N} = \{1, 2, \cdots, N\}$
Time domain channels:

- BS-k^{th} user: $\tilde{\mathbf{h}}_k^d \in \mathbb{C}^{L_d \times 1}$
- BS-n^{th} element of IRS: $\tilde{\mathbf{t}}_n \in \mathbb{C}^{L_1 \times 1}$
- IRS-k^{th} user: $\tilde{\mathbf{r}}_{k,n} \in \mathbb{C}^{L_{2,k} \times 1}$

Bandwidth divided into M sub bands
Sub bands: $\mathcal{M} = \{1, 2, \cdots, M\}$
Frequency flat within sub band

$$\tilde{\mathbf{h}}_{k,q,n} = (\tilde{\mathbf{r}}_{k,n} * \phi_{q,n}) * \tilde{\mathbf{t}}_n \in \mathbb{C}^{L_k^r \times 1}$$

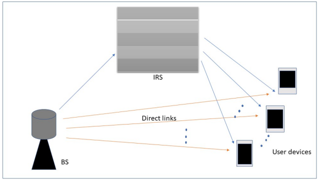


Figure: IRS assisted OFDMA DL communication

$$\mathbf{v}_{k,n} = [(\tilde{\mathbf{r}}_{k,n} * \tilde{\mathbf{t}}_n)^T, 0, 0, \cdots, 0]^T \in \mathbb{C}^{M \times 1}$$
$$\mathbf{V}_k = [\mathbf{v}_{k,1}, \mathbf{v}_{k,2}, \cdots, \mathbf{v}_{k,N}] \in \mathbb{C}^{M \times N}$$

The cascaded channel of user k:

$$\mathbf{h}^r_{k,q} = \mathbf{V}_k \phi_q \tag{1}$$

Overall channel of BS-k^{th} user: $\mathbf{h}_{k,q} = \mathbf{h}^d_k + \mathbf{V}_k \phi_q$
The CFR

$$\mathbf{H}_{k,q,m} = \mathbf{f}^H_m \mathbf{h}^d_k + \mathbf{f}^H_m \mathbf{V}_k \phi_q \tag{2}$$

$\mathbf{f}^H_m$: m^{th} row of discrete Fourier transform (DFT) matrix of size $M \times M$
$\mathbf{H}_{k,q,m}$ is perfect CSI!!

Channel estimation

Uplink pilot transmission Assumption: Orthogonal pilots

Pilot at user k, $\mathbf{x}_k \in \mathbb{R}^{M\times 1}$

K_0 out of M sub bands have non zero value,

$$K_0 = \lfloor \frac{M}{K} \rfloor$$

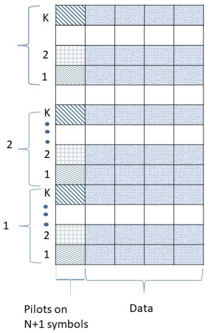


Figure: Pilot allocation scheme

$x_k^{(i)}$ being the i^{th} sub band of k^{th} user:

$$x_k^{(i)} = \begin{cases} \sqrt{p}, & \text{for} \quad i = (k_0 - 1)K + k, \quad \text{and} \quad k_0 \in [K_0] \\ 0, & \text{otherwise} \end{cases}$$

For e.g., $K = 3, M = 6$

$$\mathbf{x}_1 = [\sqrt{p}, 0, 0, \sqrt{p}, 0, 0]$$
$$\mathbf{x}_2 = [0, \sqrt{p}, 0, 0, \sqrt{p}, 0]$$
$$\mathbf{x}_3 = [0, 0, \sqrt{p}, 0, 0, \sqrt{p}]$$

$\mathbf{H}_{k,n}$ between user-k and BS (via nth element of IRS)
$\mathbf{n} \sim \mathcal{CN}(0_{M\times1}, \sigma_{BS}^2 I_{M \times M})$
The signal received at the BS

$$\mathbf{y}_{BS,n} = \sum_{k=1}^{K} \text{diag}(\mathbf{x}_k)\mathbf{H}_{k,n} + \mathbf{n} \tag{3}$$

$$\mathbf{y}_{BS,n} = \sqrt{p}\mathbf{H}_n + \mathbf{n}$$

$\mathbf{H}_n$: The equivalent channel gain between BS-users via nth element of IRS
Least squares (LS) estimate:

$$\hat{\mathbf{H}}_n = \frac{1}{\sqrt{p}}\mathbf{y}_{BS,n} \tag{4}$$

For e.g., $K = 3, M = 6$:
$\hat{\mathbf{H}}_n = \begin{bmatrix} \hat{H}_{11}^T & \hat{H}_{22}^T & \hat{H}_{33} & \hat{H}_{14}^T & \hat{H}_{25}^T & \hat{H}_{36}^T \end{bmatrix}^T$

K_0 out of M sub-band estimated for each user
Frequency interpolation to get estimation of M sub-bands

Linear interpolation:
For e.g., $K = 3, M = 6$:

$$\hat{H}_{1,2} = \frac{2}{3}\hat{H}_{1,1} + \frac{1}{3}\hat{H}_{1,4}$$
$$\hat{H}_{2,1} = \hat{H}_{2,2} - \frac{1}{3}\left(\hat{H}_{2,5} - \hat{H}_{2,2}\right)$$

Acquired: Channel estimates of single IRS element
Additional: $N + 1$ pilot symbols

Thus, estimated of M sub bands' channels for every user in particular time slot.

Downlink Communication

Sub-band allocation

$$\alpha_{k,q,m} = \begin{cases} 1, & \text{sub band } m \text{ is allocated to user } k \text{ at time slot } q \\ 0, & \text{otherwise} \end{cases}$$

$\sum_{k=1}^{K} \alpha_{k,q,m} \leqslant 1, \forall q \in \mathcal{Q}, \forall m \in \mathcal{M}$

$$\sum_{m=1}^{M} p_{q,m} \leqslant P, \forall q \in \mathcal{Q}$$

Rate of k-th user,

$$\tilde{R}_k = \frac{1}{MQ} \sum_{q=1}^{Q} \sum_{m=1}^{M} \alpha_{k,q,m} \log_2 \left(1 + \frac{\left| \hat{\mathbf{H}}_{k,q,m} \right|^2 p_{q,m}}{\Gamma \sigma^2} \right) \tag{5}$$

Noise: $\mathcal{CN}(0, \sigma^2)$ and Γ: Gap from channel capacity

The optimization problem to maximize the common (minimum) rate under perfect CSI is:

$$
\begin{aligned}
\text{(P0)} \quad & \max_{\substack{\{\alpha_{k,q,m}\},R \\ \{p_{q,m}\},\{\phi_q\}}} R && && \text{(6a)} \\
\text{s.t.} \quad & R_k \geq R, && \forall k \in \mathcal{K} && \text{(6b)} \\
& \sum_{k=1}^{K} \alpha_{k,q,m} \leq 1, && \forall q \in \mathcal{Q},\ \forall m \in \mathcal{M} && \text{(6c)} \\
& \alpha_{k,q,m} \in \{0,1\},\ \forall k \in \mathcal{K}, \forall q \in \mathcal{Q}, \forall m \in \mathcal{M} && && \text{(6d)} \\
& \sum_{m=1}^{M} p_{q,m} \leq P, && \forall q \in \mathcal{Q} && \text{(6e)} \\
& p_{q,m} \geq 0, && \forall q \in \mathcal{Q},\ \forall m \in \mathcal{M} && \text{(6f)} \\
& |\phi_{q,m}| \leq 1, && \forall q \in \mathcal{Q},\ \forall m \in \mathcal{M} && \text{(6g)}
\end{aligned}
$$

Non convex optimization problem
Difficult to solve!!!

Alternating Optimization (AO)

Solve the problem (P0) by AO; iteratively optimize $\{\alpha_{k,q,m}\}$, $\{p_{q,m}\}$ and ϕ_q

1. Optimize RB allocation

Fix $\{p_{q,m}\}$ and ϕ_q, optimize $\{\alpha_{k,q,m}\}$ by Iterative reweighted minimization (IRM) algorithm

$$
\begin{aligned}
(\mathrm{P1}) \quad & \max_{\{\alpha_{k,q,m}\}, R} \; R && && (7a)\\
\text{s.t.} \quad & R_k \geq R, && \forall k \in \mathcal{K} && (7b)\\
& \sum_{k=1}^{K} \alpha_{k,q,m} \leq 1, && \forall q \in \mathcal{Q},\ \forall m \in \mathcal{M} && (7c)\\
& \alpha_{k,q,m} \in \{0,1\},\ \forall k \in \mathcal{K}, \forall q \in \mathcal{Q}, \forall m \in \mathcal{M} && && (7d)
\end{aligned}
$$

Penalty function

$$(\tilde{\mathrm{P}}_1)\ \min_{\{\alpha_{k,q,m}\}} \sum_{m=1}^{M}\sum_{q=1}^{Q}\sum_{k=1}^{K}(\alpha_{k,q,m}+\epsilon)^r \tag{8a}$$

$$\text{s.t.}\quad \sum_{k=1}^{K}\alpha_{k,q,m}\le 1,\qquad \forall q\in\mathcal{Q},\ \forall m\in\mathcal{M} \tag{8b}$$

$$\alpha_{k,q,m}\ge 0,\quad \forall k\in\mathcal{K},\forall q\in\mathcal{Q},\forall m\in\mathcal{M} \tag{8c}$$

$$(\mathrm{P}1.1)\ \min_{\{\alpha_{k,q,m},R\}} -R+\lambda\sum_{m=1}^{M}\sum_{q=1}^{Q}\sum_{k=1}^{K}(\alpha_{k,q,m}+\epsilon)^r \tag{9a}$$

$$\text{s.t.}\quad (7b),(8b),(8c) \tag{9b}$$

$\lambda-$positive penalty parameter

IRM framework:

$$\text{(P1.2)} \min_{\{\alpha_{k,q,m},R\}} -R+\lambda r\sum_{m=1}^{M}\sum_{q=1}^{Q}\sum_{k=1}^{K} w_{k,q,m}(t)\alpha_{k,q,m} \tag{10a}$$

$$\text{s.t.} \quad (7b),(8b),(8c) \tag{10b}$$

$w_{k,q,m}(t+1) = (\alpha_{k,q,m}(t)+\epsilon)^{r-1}$
This is convex sub-problem

Use cvx for solving

2. Optimize power distribution among sub-bands

$$\text{(P2.1)} \max_{\{p_{q,m}\},R} R \tag{11a}$$

$$\text{s.t.} \quad \frac{1}{MQ}\sum_{q=1}^{Q}\sum_{m=1}^{M}\alpha_{k,q,m}\log_2\left(1+\frac{\left|\mathbf{h}_k^d+\mathbf{V}_k\phi_q\right|^2 p_{q,m}}{\Gamma\sigma^2}\right)\geq R,$$

$$\forall k \in \mathcal{K} \tag{11b}$$

$$\sum_{m=1}^{M} p_{q,m} \leq P, \qquad \forall q \in \mathcal{Q} \tag{11c}$$

$$p_{q,m} \geq 0, \qquad \forall q \in \mathcal{Q}, \forall m \in \mathcal{M} \tag{11d}$$

Convex optimization problem

Or, Use Iterative water filling algorithm

3. Optimize IRS coefficients

Fix $\{\alpha_{k,q,m}\}$ and $\{p_{q,m}\}$, then the original optimization problem (P0) can be rewritten as:

$$\text{(P3.1)} \max_{R,\{\phi_q\}} R \tag{12a}$$

$$\text{s.t.} \quad R_k \geq R, \qquad \forall k \in \mathcal{K} \tag{12b}$$

$$|\phi_{q,m}| \leq 1, \qquad \forall q \in \mathcal{Q}, \ \forall m \in \mathcal{M} \tag{12c}$$

Constraint (12b) non convex

So, successive convex approximation (SCA) to solve (P3.1)

Introducing new variables $\{y_{q,m}\}$, $\{a_{q,m}\}$ and $\{b_{q,m}\}$

$\tilde{k}(q, m) = \{k|\alpha_{k,q,m} = 1, k \in \mathcal{K}\}$

$$
\begin{aligned}
\text{(P3.2)} \quad & \max_{\substack{R,\{\phi_q\},\{y_{q,m}\}\\ \{a_{q,m}\},\{b_{q,m}\}}} R && \text{(13a)}\\
\text{s.t.} \quad & \frac{1}{MQ}\sum_{q=1}^{Q}\sum_{m=1}^{M}\alpha_{k,q,m}\log_2\left(1+\frac{y_{q,m}p_{q,m}}{\Gamma\sigma^2}\right) \geq R, && \text{(13b)}\\
& \qquad \forall k \in \mathcal{K} \\
& |\phi_{q,n}| \leq 1, \qquad \forall q \in \mathcal{Q}, \forall n \in \mathcal{N} && \text{(13c)}\\
& a_{q,m} = \Re\left\{\boldsymbol{f}_m^H\boldsymbol{h}_{d,\tilde{k}(q,m)}+\boldsymbol{f}_m^H\boldsymbol{V}_{\tilde{k}(q,m)}\phi_q\right\}, && \text{(13d)}\\
& \qquad \forall q \in \mathcal{Q}, \forall m \in \mathcal{M} \\
& b_{q,m} = \Im\left\{\boldsymbol{f}_m^H\boldsymbol{h}_{d,\tilde{k}(q,m)}+\boldsymbol{f}_m^H\boldsymbol{V}_{\tilde{k}(q,m)}\phi_q\right\}, && \text{(13e)}\\
& \qquad \forall q \in \mathcal{Q}, \forall m \in \mathcal{M} \\
& y_{q,m} = a_{q,m}^2+b_{q,m}^2, \ \forall q \in \mathcal{Q}, \forall m \in \mathcal{M}, && \text{(13f)}
\end{aligned}
$$

Still a non convex problem!!

Let $\tilde{f}(a_{q,m}, b_{q,m}) = a_{q,m}^2 + b_{q,m}^2$: Convex and differentiable.
$\tilde{f}(a_{q,m}, b_{q,m}) \geq \tilde{a}_{q,m}(2a_{q,m} - \tilde{a}_{q,m}) + \tilde{b}_{q,m}(2b_{q,m} - \tilde{b}_{q,m}) \triangleq f(a_{q,m}, b_{q,m})$

We formulate the problem (P3.2):

$$\text{(P3.3)} \quad \max_{\substack{R,\{\phi_q\},\{y_{q,m}\} \\ \{a_{q,m}\},\{b_{q,m}\}}} R \tag{14}$$

$$(13\text{b}), (13\text{c}),$$
$$(13\text{d}), (13\text{e})$$
$$y_{q,m} \geq f(a_{q,m}, b_{q,m}) \tag{15}$$

This is convex optimization problem. Solve using CVX
Update $\tilde{a}_{q,m}, \tilde{b}_{q,m}$ based on solution to (P3.3)

Algorithm

Input: $\{\mathbf{h}_k^d\}, \{\mathbf{V}_k\}, P, K, N, M, Q, \Gamma, \sigma^2, I$
Output: $\{\alpha_{k,q,m}^*\}, \{p_{q,m}^*\}, \{\phi_q^*\}, R^*$
for $i = 1$ **to** I
  Randomly generate $\phi_{q,m}$ with uniform phase distribution for each $\phi_{q,m}$
  repeat
    Optimize $\alpha_{k,q,m}$ via IRM algorithm.
    Optimize $p_{q,m}$ simply via cvx.
    Optimize ϕ_q via SCA.
  until R converges to a prescribed accuracy
  Record the optimal solution as $\left(\{\alpha_{k,q,m}^{(i)}\}, \{p_{q,m}^{(i)}\}, \{\phi_q^{(i)}\}, R^{(i)}\right)$
end for
Set $i^* = \max_{i=1,2,\cdots,I} \mathrm{R}^{(i)}, \left(\{\alpha_{k,q,m}^*\}, \{p_{q,m}^*\}, \{\phi_q^*\}\right) \leftarrow \left(\{\alpha_{k,q,m}^{(i*)}\}, \{p_{q,m}^{(i*)}\}, \{\phi_q^{(i*)}\}\right)$

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